

Minerals

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Research Topic

I decided to research the global lithium production from 2019-2023. Lithium is a key mineral for most energy transition. As the world is becoming more and more energy dependent, analyzing lithium production and mining is important for future energy use (like EV adoption). This primarily covers how much lithium is being mined every year, and which countries are mining.

Code

```
library(ggplot2)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter, lag`

The following objects are masked from 'package:base':

`intersect, setdiff, setequal, union`

```
library(scales)
```

```
# --- Data (metric tons of lithium content) ---
```

```

df <- data.frame(
  country = rep(c("Australia","Chile","China","Argentina",
                  "Brazil","Zimbabwe","Portugal"), each = 5),
  year    = rep(2019:2023, times = 7),
  production_t = c(
    45000, 39700, 55300, 74000, 86000, # Australia
    19300, 21500, 28300, 39000, 44000, # Chile
    10800, 13100, 14000, 19000, 33000, # China
    6300,  5900,  5970,  6200,  9600, # Argentina
    300,  1420,  1500,  2200,  4900, # Brazil
    1200,  417,   712,   800,  3400, # Zimbabwe
    1200,  348,   900,   600,   380  # Portugal
  )
)

# Order countries by total production so the stack reads largest -> smallest
country_order <- df %>%
  group_by(country) %>%
  summarise(total = sum(production_t), .groups = "drop") %>%
  arrange(desc(total)) %>%
  pull(country)

df$country <- factor(df$country, levels = rev(country_order))

# Colorblind-safe Okabe-Ito palette
palette <- c(
  "Australia" = "#0072B2", "Chile"      = "#D55E00",
  "China"      = "#E69F00", "Argentina" = "#009E73",
  "Brazil"     = "#CC79A7", "Zimbabwe"  = "#56B4E9",
  "Portugal"   = "#999999"
)

# --- Plot ---
p <- ggplot(df, aes(x = year, y = production_t, fill = country)) +
  geom_area(alpha = 0.95, color = "white", linewidth = 0.3) +
  scale_fill_manual(values = palette, breaks = country_order) +
  scale_y_continuous(labels = label_comma(),
                     expand = expansion(mult = c(0, 0.05))) +
  scale_x_continuous(breaks = 2019:2023) +

```

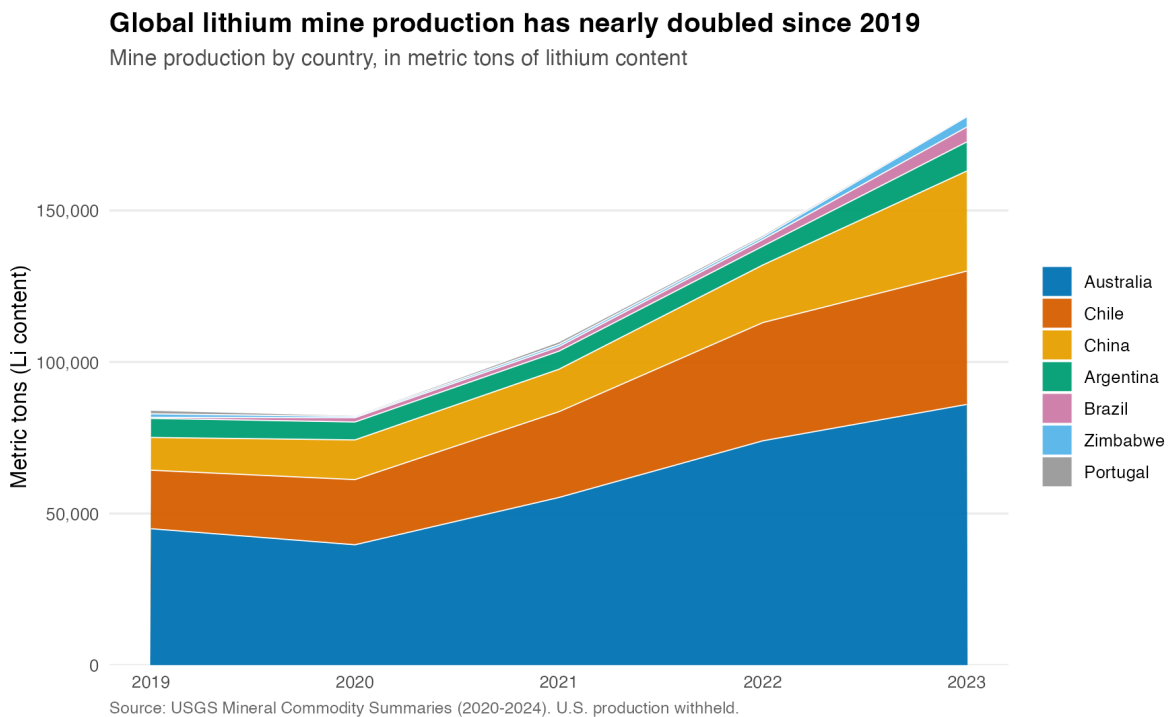
```

labs(
  title      = "Global lithium mine production has nearly doubled since 2019",
  subtitle   = "Mine production by country, in metric tons of lithium content",
  x = NULL, y = "Metric tons (Li content)", fill = NULL,
  caption    = "Source: USGS Mineral Commodity Summaries (2020-2024). U.S. production withheld."
) +
theme_minimal(base_size = 12) +
theme(
  plot.title      = element_text(face = "bold", size = 15),
  plot.subtitle   = element_text(color = "grey30", margin = margin(b = 10)),
  plot.caption    = element_text(color = "grey40", hjust = 0, size = 9),
  panel.grid.minor = element_blank(),
  panel.grid.major.x = element_blank(),
  legend.position = "right"
)

ggsave("lithium_chart.png", p, width = 9, height = 5.5, dpi = 200, bg = "white")

```

Graph



The graph along with the source material included in this report, show that lithium production has doubled between 2019-2023. Lithium production has risen from 4,000 to 181,000 metric tons of contained lithium, driven by surging demand for EV and grid-storage batteries. Australia accounts for around half of the world's output of lithium. Chile and China have 2nd and 3rd most lithium production. Argentina and Brazil are smaller, but are definitely growing in mining. Respectively, Zimbabwe and Portugal remain at a smaller production amount. The world is mining lithium more than it was before 2019, and will continue to mine more to accommodate the demand for this mineral in everyday tech uses.